

Пробная олимпиада ISTCiM – 2

Time allowed: 4 hours

14 сентября 2026 года

A. Algebra and combinatorics. Prove that if p, q and r are three distinct primes, then every group of order pqr is solvable.

C. Calculus and mathematical analysis. For a positive integer n let $\langle n \rangle$ denote the integer closest to $\sqrt{n}$. Evaluate

$$\sum_{n=1}^{\infty} \frac{2^{\langle n \rangle} + 2^{-\langle n \rangle}}{2^n}.$$

E. Equations and inequalities. Let $f: [-1, 1] \rightarrow \mathbb{R}$ be continuous and such that

$$f(x) = \frac{2-x^2}{2} f\left(\frac{x^2}{2-x^2}\right) \quad \text{for all } x \in [-1, 1], \quad f(0) = 1,$$

and such that $\lim_{x \rightarrow 1^-} \frac{f(x)}{\sqrt{1-x}}$ exists and is finite. Prove that f is uniquely determined by these conditions and find it.

G. Geometry and linear algebra. Let $A, B \in M_n(\mathbb{C})$ satisfy $A^2 + B^2 + A - B = 2(AB + I_n)$. Prove that

$$\operatorname{tr}((A - B)(A - B + I_n)) = 2n \quad \text{and} \quad \det((A - B)(A - B + I_n)) = 2^n.$$

P. Set theory and probability. An integer n , unknown to you, is chosen uniformly at random in $[1, 2002]$. Your aim is to name n on an odd-numbered guess. After each wrong guess you are told whether n is larger or smaller, and your next guess must be one of the integers still possible. Show that you have a strategy which succeeds with probability more than $2/3$.